

Advantages and Disadvantages of Modern Language Models

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Abstract—Large language models (LLMs) have quickly grown to include trillions of parameters and million-token context windows, leading to big improvements in performance on benchmark tasks. But when these models are used in real-world settings, they face challenges like limited memory bandwidth, slow response times, high energy use, and difficulties in scaling infrastructure. These issues show that just adding more parameters won't be enough for continued progress. Instead, we need to focus on making model designs more efficient, considering physical, economic, and environmental limits. This study compares two types of model designs—retrieval-augmented systems and extended-context transformers—under realistic conditions. We use a unified evaluation method based on Token Economics to measure how efficient a model is by looking at token throughput, memory transfer costs, and energy-normalized latency. We tested models ranging from 7B to 70B parameters with context windows up to 1 million tokens, using both static reasoning and dynamic knowledge workloads. The results show that retrieval-augmented designs reduce inference costs significantly for dynamic knowledge tasks and perform better in environments with limited bandwidth. On the other hand, extended-context transformers face issues because their attention mechanisms and KV-cache grow in a way that leads to slower performance. Looking at the infrastructure, we found that factors like interconnect bandwidth, rack power density, and heat management are major limits to scaling. These findings support moving toward hybrid, retrieval-aware, and memory-efficient models as more sustainable options compared to just scaling up models. This approach helps reduce environmental impact and lowers the risks of large-scale AI deployment.

Index Terms — *Token Economics, Retrieval Augmented Generation, Memory Bound Computing, Infrastructure Scalability, Hardware Constraints.*

I. INTRODUCTION

Artificial intelligence has made huge progress, especially with the rise of large language models (LLMs). These models have changed how computers handle natural language, like understanding, creating, and using it in new ways. Since the Transformer structure was introduced in 2017, LLMs have grown from having just millions of parameters to hundreds of billions, and some recent models now have trillions [1]. These models can do many different things, such as solving difficult problems, writing code, creating stories, and even helping with scientific research.

The fast development from GPT-3's 175 billion parameters to models that can handle over a million words at once shows a major technological leap [2].

This growth has allowed LLMs to read whole books, have long and meaningful conversations, and do complex thinking that wasn't possible before. As these models are used in areas like healthcare, education, research, and business, they have caused both excitement and worry among experts and professionals.

However, the aggressive scaling paradigm faces increasing physical and economic constraints. The exponential growth in model parameters, context windows, and computational demands has led to significant challenges in memory bandwidth, inference latency, power consumption, and infrastructure scalability [3]. Recent studies show that practical deployment increasingly reveals fundamental limitations that pure parameter expansion alone cannot overcome. Memory bandwidth limitations, rising energy costs, and hardware constraints indicate that continued progress requires architectural innovation rather than simple scaling [4].

At the same time, the rapid commercialization and wide deployment of autonomous AI agents across digital ecosystems highlight the systemic impact of architectural design decisions.

Healthcare systems using LLM-based diagnostic tools, educational platforms employing AI tutors, and enterprise systems implementing automated decision-making agents all rely heavily on the efficiency, reliability, and sustainability of underlying architectures [5, 6]. The environmental impact of large-scale LLM deployment, particularly energy use and carbon footprint, further complicates the sustainability of current scaling trends.

The contrast between scaling achievements and practical deployment challenges requires careful assessment of architectural alternatives.

Two main paradigms have emerged: retrieval-augmented generation (RAG) systems that utilize external knowledge bases to reduce model size, and extended-context transformers that enhance performance through enlarged internal capacity [7, 8]. Each approach presents unique benefits and drawbacks that must be evaluated within real-world deployment contexts.

This research addresses the urgent need for a thorough evaluation of these architectural trade-offs.

By introducing a unified framework based on Token Economics, we systematically examine the efficiency, scalability, and sustainability implications of different architectural choices. The contributions of this work include: (1) a comprehensive framework for assessing LLM architectures under realistic deployment constraints, (2) a systematic comparison of retrieval-augmented and extended-context approaches across multiple dimensions, (3) an infrastructure-level analysis identifying key bottlenecks and scaling limits, and (4) recommendations for sustainable architectural design that balance performance, efficiency, and environmental responsibility.

II. BACKGROUND AND LITERATURE REVIEW

A. Evolution of Large Language Models

The Transformer architecture, introduced by Vaswani and others [1], laid the groundwork for today's large language models. It uses a self-attention mechanism and allows parallel processing, which is a big improvement over the step-by-step way recurrent neural networks work. This change makes training models faster and easier to scale up. The attention mechanism helps the model focus on important parts of the input, creating a better understanding of the context by considering both nearby and distant parts of the text.

Later improvements have greatly increased the size and power of these models.

Models like BERT, GPT-2, and GPT-3 showed that making models bigger and training them on more data can lead to new abilities that weren't planned during training [2]. Research by Kaplan and others [9] explained why increasing model size, data, and computing power usually leads to better performance.

Newer models have gone even further by tackling the problem of handling very long texts.

The standard attention method has a problem with speed and memory when dealing with long sequences because it grows quickly with the length of the text. To solve this, new designs like Transformer-XL [10], Longformer [11], and Infini-attention [12] have been created. These changes allow models to process texts that are thousands or even millions of words long.

B. Retrieval-Augmented Generation

Retrieval-Augmented Generation is a different way of storing and accessing knowledge. Instead of putting all the information inside the model, RAG systems get the needed information from outside databases when the model is used [7]. This method has some good benefits: it uses smaller models, needs less computing power when updating knowledge, and can get up-to-date information that wasn't available when the model was first made.

Guu et al. [13] found that using retrieval methods can greatly improve how well language models work, while also needing fewer parameters.

Models like REALM and others showed that adding knowledge retrieval to language models helps them perform better in tasks like answering open-ended questions and solving complex problems. Lewis et al. [14] also showed that RAG models can do well in several tests while using much fewer parameters than other models that rely only on their own parameters. The way RAG systems are built also makes it easier to adapt them to different fields without having to retrain the whole model, which cuts down on the computing power needed for special uses..

C. Extended-Context Architectures

Extended-context transformers use a different approach, increasing their internal ability to handle longer sequences without needing extra retrieval methods. Dai et al. [10] created Transformer-XL, which uses recurrence at the segment level, allowing models to understand longer relationships while keeping computations efficient. Longformer and other similar designs use sparse attention patterns to

reduce computational requirements while preserving the ability to process extended sequences [11, 15].

Recent improvements have made it possible to handle very long sequences of text. Ring Attention [16] allows processing of sequences with more than 500,000 tokens by using distributed computing. Another approach, Infini-attention, shows that by efficiently using memory, sequences can be up to 1 million tokens long without using too much memory [12]. But, even with these advances, there are big challenges when dealing with longer sequences: as the sequence gets longer, the memory needed to store key-value pairs (KV cache) increases, and the attention mechanism's performance depends on the square of the sequence length, which makes the computation much more expensive as the sequence gets longer [17].

A big breakthrough came with FlashAttention [15], which changes how attention is calculated to avoid storing the full $N \times N$ attention matrix in slow memory (HBM).

Normally, standard attention requires the following number of HBM accesses:

$$HBM \text{ accesses (standard)} = \Theta(Nd + N^2) \quad (1)$$

FlashAttention reduces this to:

$$HBM \text{ accesses (FlashAttention)} = \Theta(N^2 d^2 M^{-1}) \quad (2)$$

where N is the sequence length, d is the head dimension, and M is the size of fast on-chip memory (SRAM), leading to up to 9 times fewer HBM accesses. FlashAttention uses a technique that keeps intermediate results in fast SRAM and breaks down the softmax calculation into smaller parts, allowing it to use memory in a way that grows proportionally with the sequence length. In practice, FlashAttention speeds up GPT-2 by 3 times for 1K token sequences, improves BERT-large by 15%, and uses memory more efficiently than traditional methods, making it possible to run tasks like Path-X (16K tokens) and Path-256 (64K tokens) [15].

Ring Attention [16] solves the problem of communication delays between devices by imagining multiple devices as a ring.

It overlaps the transfer of KV blocks between devices with the computation that happens on each device. If the time needed for each block of computation is longer than the time needed to transfer data, there's no extra delay. The length of the sequence can increase as the number of devices increases, and each device only needs to handle a limited amount of data, regardless of the total sequence length. On a TPUv4-1024 cluster, this allows for training on sequences longer than 100 million tokens [16].

D. Deployment Challenges and Infrastructure

Constraints

When LLMs are used in real situations, many issues become clear that aren't seen in tests. Pope et al. [18] looked at the costs of running models and found that memory bandwidth, not just computing power, is often the main problem. Patterson et al. [19] studied how much carbon is used to train big models and discovered that the carbon emissions can be as much as traveling across the world several times by plane. Other hardware limits like memory size, how fast data can move between parts, power use, and heat also affect how models are designed and how well they work in real life..

E. Quality and Reliability Challenges

Besides problems with the basic setup, large language models also have issues with their quality and reliability, which can make it hard to use them in real situations. New studies show that these models often create wrong or made-up references, and in some cases, up to 60% of the citations they

produce are fake [21]. To measure these problems, Aljamaan and others created the Reference Hallucination Score (RHS), which shows that models like ChatGPT and Bing often make serious mistakes when generating citations.

Looking at 4,900 summaries about science that were written by large language models, we found that these models often make overly broad statements. This happened in 26 to 73% of the cases for models like DeepSeek, ChatGPT-4o, and LLaMA 3.3 70B [23]. The summaries from these models are almost five times more likely to make general statements compared to summaries written by humans (odds ratio = 4.85, $p < 0.001$) [23]. This is especially concerning in areas like healthcare education, where it could harm academic honesty and spread false information [24].

III. METHODOLOGY

A. Evaluation Framework: Token Economics

This research presents a Token Economics framework that helps evaluate large language models (LLMs) in a structured way, especially when considering real-world limitations. The framework looks at how efficiently a model can process tokens by considering several factors like how many tokens can be handled at once, the cost of moving data between different memory levels, and the energy used per token, adjusted for energy use. It treats the token, which is the basic unit of computation in LLMs, as the central element for analyzing efficiency. The framework looks at all the costs involved in creating and processing each token..

The framework breaks down the cost of generating each token into five parts: (1) Token Processing Cost: the amount of computation needed for each token, measured in FLOPs; (2) Memory Transfer Cost: the number of bytes moved between different memory levels for each token; (3) Context Maintenance Cost: the memory and computation needed to keep track of the sequence as it's being built; (4) Retrieval Cost (for systems that use external knowledge): the computational and memory resources used to search for and combine information from outside sources; and (5) Energy Cost: the total amount of energy used to generate each token, including computing, memory access, and other system activities. This breakdown helps identify specific strengths and weaknesses of different designs, going beyond overall performance numbers to show the real trade-offs in efficiency..

B. Experimental Design

Comparative experiments assess architectural decisions across various dimensions. Model scale varies from 7B, 13B, 30B, and 70B parameters, covering both edge deployment (7B-13B) and data-center scenarios (30B-70B). Context window size ranges from 4K to 1M tokens. Two main workload types are analyzed: Static Reasoning, which involves logical, mathematical, or code generation tasks where all necessary information is included in the query, and Dynamic Knowledge, which requires access to current or domain-specific data not present in the training set.

RAG experiments test different things like how many documents are used for retrieval, which can range from 1 million to 100 million. They also look at the type of system used to find information, such as dense models versus sparse models. Another thing they check is how many results are returned, from the top 5 to the top 100. When it comes to deploying these systems, they look at three setups: one uses powerful cloud servers with high-performance GPUs like NVIDIA A100 or H100 and fast connections between them. Another setup uses regular hardware that has less memory and is meant for use on the edge, like at the edge of a network.

The third setup tries to mimic situations where the system starts up without being always on, which is common in serverless environments and needs special handling when it first starts.

C. Measurement Methodology

Experiments gather detailed measurements in three areas. Performance Metrics cover how well tasks are done on MMLU, HellaSwag, and other specific benchmark tests; how good the responses are based on human judgment and computer-based scores; and how long tasks take, measured for different lengths of text and different group sizes. Efficiency Metrics are also included. encompass memory utilization profiling including peak memory, KV-cache size, and memory bandwidth utilization; and energy consumption measured through hardware power sensors. Infrastructure Metrics cover scaling behavior across distributed inference configurations, thermal and power density profiles for rack-level deployment, and interconnect bandwidth utilization.

D. Analysis Methods

Statistical analysis uses generalized linear mixed models to handle repeated measurements across various model setups and workloads, allowing for thorough evaluation of architectural impacts while adjusting for confounding factors. Cost-benefit analysis evaluates the trade-offs between performance improvements and resource needs, taking into account both initial deployment costs and ongoing operational expenses. Scaling analysis examines how costs and benefits change with model size, context length, and deployment scale, forecasting behavior beyond the tested configurations

E. IO-Complexity Metrics and AI Engagement Modalities

To include Flash Attention in the Token Economics framework, we add IO-complexity analysis along with FLOPs to the Memory Transfer Cost part. As shown in equations (1) and (2), regular attention uses $\Theta(Nd + N^2)$ HBM accesses, but FlashAttention lowers this to $\Theta(N^2d^2M^{-1})$, where M represents on-chip SRAM. This measure is checked through experiments with different context lengths to show how much memory transfer is saved per token.

We also look at two different ways AI affects human work.

The first is Active Engagement, where AI helps directly, like in the GitHub Copilot study [25], which involved 95 professional programmers on Upwork doing a standard HTTP-server task, timed from when they created a repository to when they made the first commit that passed a 12-check test. The second is Passive Exposure, where people come across AI-generated ideas without getting direct help, as seen in the dynamic Alternate Uses Task study [28], which had 844 participants from 48 countries. Comparing these two types shows how LLMs impact both professional productivity and creative thinking. .

IV. RESULTS

A. Static Reasoning Performance

Extended-context transformers show consistent benefits in static reasoning tasks where all required information is included in the query. The 13B extended-context model performs similarly to the 70B standard-context model on reasoning benchmarks, indicating that increasing context length can be an alternative to parameter expansion in certain situations. However, this performance gain comes at a significant memory cost: KV-cache requirements grow linearly with context length, leading to deployment limitations. Initial performance improvements are notable as context expands from 4K to 32K tokens, offering enough

capacity for complex reasoning processes. Yet, improvements become minimal beyond 100K tokens, with small gains needing exponential memory increases. For 1M token contexts, the KV-cache reaches 150GB with 70B models, surpassing available GPU memory even in high-end setups. Peak memory bandwidth usage reaches 90% during continuous generation, confirming that memory transfer, not computation, limits throughput.

B. Dynamic Knowledge Task Performance

RAG architectures show significant benefits in handling dynamic knowledge tasks. The 7B RAG model achieves accuracy within 2% of the 70B standard model on knowledge-intensive benchmarks, while using 10 times fewer parameters and 90% less memory during inference. RAG systems are particularly advantageous for temporal queries needing current data, domain-specific knowledge requiring specialized databases, long-tail facts that are rare in training data, and multi-source synthesis tasks. However, they have drawbacks in tasks that demand deep parametric knowledge integration, situations with poor retrieval accuracy, and applications sensitive to retrieval latency. Retrieval costs are generally manageable for most applications. Dense retrieval over a 100M document corpus takes 10-20ms per query, accounting for 5-10% of the end-to-end latency for 7B models. This overhead decreases proportionally for larger models where generation latency is dominant, and retrieval latency can be parallelized with initial token generation to further reduce perceived overhead.

C. Memory and Bandwidth Analysis

Memory bandwidth becomes the main constraint for extended-context architectures. Standard setups achieve 60-70% peak bandwidth utilization, whereas extended-context configurations reach 85-95%, indicating saturation that leads to latency variation and throughput instability, which is unsuitable for production use. KV-cache expansion adds more pressure on the memory system, with memory growing linearly per attention head and multiplied by batch size in batched inference scenarios. Memory-efficient attention mechanisms offer partial relief: Flash Attention and similar methods optimize memory access patterns but do not tackle the fundamental storage needs, offering only 2-3x efficiency gains. FlashAttention [15] demonstrates IO-aware efficiency improvements. By tiling attention into SRAM-resident blocks and applying selective recomputation during the backward pass—storing outputs and softmax statistics instead of the full $N \times N$ attention matrix—it achieves up to $9 \times$ fewer HBM accesses than standard attention. In practice, this provides a $3 \times$ speedup on GPT-2 (1K sequence length), a 15% improvement on BERT-large, and up to $20 \times$ better memory efficiency, enabling Path-X (16K) and Path-256 (64K) benchmarks that were previously out of computational reach..

D. Energy Efficiency Analysis

Extended-context models use 3 to 5 times more energy for each token in long sequences compared to RAG methods. This happens because of how memory is used: accessing data from DRAM uses about 100 times more energy per bit than using cache, and 1000 times more than using registers. Extended-context models depend on big KV-caches stored in DRAM, which leads to a lot of extra energy use that RAG systems don't have. RAG systems are 10 times more energy efficient than pure extended-context models when handling 1 million tokens, and they still perform just as well. When you look at the whole system, like cooling, power delivery, and facility costs, these energy differences get even bigger, especially when using these systems at very large scales.

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E. Infrastructure Scalability

Infrastructure analysis shows three main problems. First, Interconnect Bandwidth: when doing distributed inference, you need to move KV-caches between devices. Models that can handle longer text (extended-context models) need 5 to 10 times more bandwidth than other methods like RAG for the same model size. Second, Power Density: a rack with 8 GPUs running a 70B model uses 40 to 60 kW of power. These models push power use past what most data centers can handle, which is usually 20 to 25 kW per rack. Third, Thermal Constraints: high-density setups create a lot of heat in one place, so you need special cooling systems. This makes deployment more expensive and limits where you can put these systems. RAG architectures are better for scaling infrastructure because they use smaller models that can run on regular hardware and allow retrieval systems to be spread out across storage and computing parts.

Ring Attention helps solve the interconnect problem by overlapping the transfer of KV-blocks with the calculation of attention in blocks.

On a 32-node A100 cluster with a 7B model, the maximum context length goes from 128,000 to 4,096,000 tokens. On a TPUv4-1024 cluster with a 3B model, it goes from 32,000 to 16,384,000 tokens, and the model's computing efficiency stays good. In the ExoRL reinforcement learning test, Ring Attention lets you use 128 different paths (about 64,000 tokens) at the same time, which was previously limited to 32. A LLaMA-13B model trained with Ring Attention performs well on line-retrieval tests with 512,000 tokens, and does much better than Claude-2 and GPT-3.5 at that level [16].

F. Hybrid Architecture Performance

Hybrid architectures that integrate moderate context expansion (32K-100K tokens) with RAG mechanisms show better overall efficiency. The 13B hybrid setup (32K context plus RAG) achieves 95% of extended-context performance on reasoning tasks, 98% of RAG performance on knowledge tasks, uses 50% of extended-context memory, and offers 80% better energy efficiency compared to pure alternatives. Memory needs fit within single GPU setups, avoiding the complexity of distributed inference, and energy use allows for scalable, sustainable deployment.

G. Real-World Productivity and Accessibility Benefits

Empirical studies show real evidence that using large language models (LLMs) improves productivity. Research on GitHub Copilot found that developers who use AI assistance finish coding tasks 55.8% faster than those who don't, with a 95% confidence interval between 21% and 89% [25]. In a controlled experiment with 95 professional programmers, the group using AI completed an HTTP server setup in 71.17 minutes, while the group without AI took 160.89 minutes, again with the same confidence interval. Interestingly, both groups thought the productivity gain would only be about 35%, which is much lower than the actual improvement. When asked how much they would pay for the AI tool, the group using it said they'd pay an average of \$27.25 per month, compared to \$16.91 for those not using it, showing they see real value beyond just speed. Further analysis shows that developers with less experience, those who code a lot every day, and older programmers (aged 25-44) benefited the most. This suggests AI tools could help people move into software development and make the field more accessible. If this 55.8% productivity gain spreads across jobs in computer and math fields, it could have a big impact on the economy and save a lot of money worldwide [25].

A study called the Passive Exposure study used the Alternate Uses Task (AUT) with 844 people from 48 countries [28]. When people were exposed to ideas made by ChatGPT, the variety of ideas created as a group went up a lot (Cliff's delta = 0.31, $p = 0.001$), but the creativity of individual ideas didn't improve ($p = 0.97$). This shows that AI helps produce more varied ideas, but doesn't make individual ideas better. Ideas from ChatGPT-3.5 were seen as more original than the average person's ideas (beta = 0.62 increase in originality). Using AI also made the group ideas become more varied faster, stopping the usual trend where people in groups start to think alike. When it came to using ideas labeled as AI-generated, people with lower confidence in their own creativity were less likely to use them, especially for easy tasks ($\rho = 0.3$) but more likely for harder tasks ($\rho = 0.8$). However, highly creative people used AI ideas based on how good the content was, not whether it was labeled as AI.

Large Language Models (LLMs) are also very good at handling many languages.

Studies show they can translate complex scientific articles accurately in 28 languages, keeping the original JATS XML format [26]. The quality of translations stays high, ranging from 91.7% to 98.0% across different languages. Scientists found that 93.3% of the translated papers were accurate, which could help people who don't speak English access scientific information around the world.

H. Economic Exposure and Labour-Market Implications

The Anthropic Economic Index [29] measures something called Observed Exposure, which looks at how much real-world use there is of large language models (LLMs) in different tasks. It considers both how capable the models are in theory and how much they are actually used, with some tasks weighted more heavily than others. For example, automated API calls are considered more important than chat-based interactions.

A big finding is that there's a big difference between what the models could do in theory and what they are actually doing in practice. About 94% of tasks in Computer and Mathematics jobs are theoretically possible for LLMs, but only about 33% are actually being used now. This gap is because of things like poor infrastructure, rules and regulations, and how well these workflows are adopted in the real world.

Some of the most exposed jobs include Computer Programmers (75%), Customer Service Representatives (71%), Data Entry Keyers (67%), and Financial Analysts (61%). People in these high-exposure roles are usually older, more likely to be women, more educated, and earn more than those in low-exposure roles. Even though these jobs are highly exposed to AI, there hasn't been a big increase in unemployment for people in these roles up to early 2026.

However, there are signs of a "hiring chill"—meaning fewer young people are entering these high-exposure jobs. Specifically, workers aged 22 to 25 entering these fields dropped by about 14% in 2024 compared to 2022. This is raising concerns about a "junior talent gap" in areas like software engineering and finance. According to the Bureau of Labor Statistics, every 10% increase in AI coverage is linked to a 0.6% decrease in the projected growth of these jobs by 2034. [29].

V. DISCUSSION

A. Implications for Architectural Design

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B. Environmental and Sustainability Considerations

Looking at how much energy is used, we see big effects on the environment. Moving to more efficient system designs can cut the carbon footprint of large language models by 50 to 80 percent without lowering their performance. To be environmentally friendly, we also need to look at the total cost over the whole life of these systems, including the energy used to make the GPUs and the materials that are hard to get. These findings suggest that policies like carbon taxes that support efficient systems, rules that require energy usage to be open and clear, and government buying practices that push for greener options can all help speed up the move to more sustainable ways of doing things.

C. Technical Advances in Memory-Efficient Architectures

The Infini-attention mechanism achieves a 114x compression ratio compared to Memorizing Transformers, while maintaining bounded memory usage for unlimited context [12]. This approach integrates compressive memory into standard attention, addressing the fundamental limitation of standard attention, which scales quadratically with 3TB memory requirements for 500B models at 2048 context windows. Ring Attention further demonstrates that blockwise computation of self-attention distributed across multiple devices can train sequences exceeding 100 million tokens without approximation, achieving 512x improvement over previous state-of-the-art memory-efficient transformers and enabling 16M token context for 3B models [27].

Combining FlashAttention's per-device IO-awareness with Ring Attention's ring-topology distribution yields a compounding effect: FlashAttention reduces per-device HBM pressure from $O(N^2)$ to $O(N^2 d^2 M^{-1})$ accesses, while Ring Attention distributes the remaining KV-cache burden linearly across devices.

Together, they represent a hardware-algorithm co-design pathway capable of extending context capability to new modalities including high-resolution long-form video, full-

codebase understanding, and complex scientific dataset analysis, without proportional growth in per-device memory or energy consumption.

D. Quality, Reliability, and Systemic Risk

RAG systems have a natural benefit for making sure citations are accurate because they use content that can be checked against real sources. Elicit and SciSpace have very little hallucination because they rely on retrieval methods, while models that only generate text show much higher rates of making up information [22]. The tendency of large language models to generalize too much in scientific summaries, along with problems in healthcare education like threats to academic honesty and the spread of false information [24], add to the challenges in deploying these systems. This shows that just making models more efficient in terms of their structure isn't enough to solve all the problems when putting them into use.

Using AI agents on a large scale creates risks that can get worse depending on how the systems are built.

Architectures that need a lot of computing power make it easier for control and capability to be concentrated in organizations with lots of money. RAG and mixed systems help by using fewer resources, which allows for a wider variety of uses and lowers the chance of a single point of failure. Systems that are efficient and spread out also make the whole setup more reliable, even if part of the system has a problem.

E. Economic Sustainability and the Observed-Exposure Gap

Comparing the high energy and computing costs mentioned in Section IV-D with the observed exposure results from [29] brings up an important question about return on investment: right now, the money spent on large language model infrastructure isn't leading to the automation abilities we expect. The big gap between what's possible and what's actually happening, like the 94% versus 33% difference in Computer and Math jobs, shows that things like infrastructure, rules, and how work flows are holding things back. This affects both the economic benefits and the risks of job loss. The 14% drop in hiring for people aged 22-25 in fields that are heavily affected by automation suggests a bigger change is coming. If the actual exposure gets closer to what's possible without making the systems more efficient, the environmental cost of all this automation could be really high. This supports the paper's main point that making systems more efficient, not just making them more powerful, is key to making AI use both economically and environmentally sustainable.

F. Limitations and Future Directions

This study has some limits. It doesn't cover the newest models that have trillions of parameters, and when we try to guess what might happen with bigger models, we need to be careful. New hardware is improving quickly, so some of the current limits might not be as tight in the future. New types of model designs could change how cost and benefits compare, and looking at specific uses might show different things when models are used in special ways. Future work could focus on: (1) Flexible model designs that switch between using extra knowledge and longer text understanding depending on the task; (2) Creating special hardware that works well with these new models; (3) Studying the bigger picture effects by looking closely at risks and benefits; and (4) Long-term studies that look at the whole life of these models from start to finish.

VI. CONCLUSION

This study looks closely at the pros and cons of different types of modern language model designs when they are used

in real-world situations. Using the Token Economics framework, we look at two main types of models: those that use extended context and those that use retrieval methods. We find that there are key trade-offs that must be understood for these models to develop in a sustainable way.

Extended-context models, which can handle longer texts, work well for tasks that need steady, focused thinking.

However, they come with several big challenges. They use a lot more memory, their processing speed slows down quickly as the text gets longer, they use a lot of memory bandwidth, and they use a lot more energy than other types of models. In fact, they use 3 to 5 times more energy than more efficient options.

On the other hand, retrieval-based models do better when the information needed is changing or not fixed.

These models are much more energy-efficient, which means they have a much smaller environmental impact—up to 80% less than the extended-context models.

The research suggests that the best approach is to use a mix of both types.

By expanding the context to a moderate size (between 32,000 and 100,000 words) and adding retrieval methods, the models can perform almost as well as the pure extended-context models, but with much better efficiency.

In addition to efficiency, language models have other practical benefits too.

For instance, they can help improve software development tasks by 55.8% and can translate scientific texts accurately across 28 languages with 95.9% accuracy. New methods like Infini-attention and Ring Attention show that it's possible to greatly reduce the amount of memory used, which means that the main limit to model performance might not be the model design itself, but the hardware and infrastructure it runs on.

The growth of large language models isn't just about making them bigger, but also about making them more efficient and better suited for how they are used.

The choices made in model design affect the overall system and the way AI is used. Efficient models can be used in many different places instead of just in one big data center, which helps reduce the pressure to centralize everything and supports environmental sustainability. To fully benefit from these models, it's important to design them in a responsible way that takes into account performance, efficiency, and the impact on the larger system..

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VIII. REFERENCES

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